

Self-Generating and Self-Renewing Zwitterionic Polymer Surfaces for Marine Anti-Biofouling

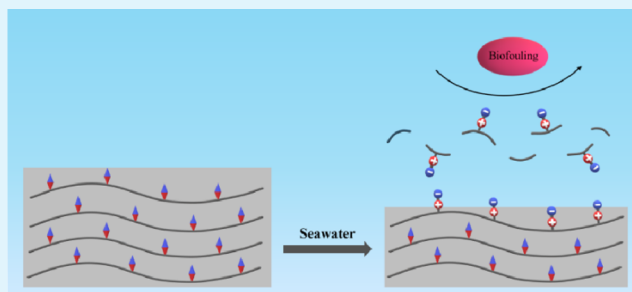
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S Supporting Information

ABSTRACT: Regeneration of antifouling polymer surfaces after contamination or damage is an important issue, especially in complex marine environments. Here, inspired by the self-renewal of silyl acrylate polymers and the protein resistance of zwitterionic polymers, we prepared a novel hydrolysis-induced zwitterionic monomer, tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSA), and copolymerized it with methyl methacrylate (MMA). Such a copolymer rapidly self-generates a zwitterionic surface and provides fouling resistance in marine environments. Furthermore, TCBSA was copolymerized with MMA and 2-methylene-1,3-dioxepane (MDO), where MDO causes degradation of the polymers. Our study demonstrates that the degradation of the polymer is controlled, and the degradation rate increases with the external enzyme concentration in the seawater, leading to a self-renewing dynamic surface. Quartz crystal microbalance with dissipation measurements show that the polymeric coating with self-generating zwitterions has excellent protein resistance in seawater. Bioassays demonstrate that the coating can effectively inhibit the adhesion of marine bacteria (*Pseudomonas* sp.) and diatoms (*Navicula incerta*). The coating with a self-generating and self-renewing zwitterionic surface is potential to find applications in marine anti-biofouling.

KEYWORDS: antifouling coating, degradable polymer, protein resistance, zwitterionic surface, radical ring-opening polymerization



INTRODUCTION

Marine biofouling that has adverse effects on marine activities is a sticky problem.^{1–3} Using antifouling coatings is the most convenient and economical method to prevent biofouling. Generally, they are designed to act in three ways including antifoulant releasing, fouling resistance, and fouling release.^{4,5} The antifoulant-releasing or fouling resistance coatings can actively prevent the attachment or growth of marine organisms on submerged surfaces. The surfaces of fouling release coatings cannot prevent the attachment of fouling organisms, but the adhesion between organisms and the surfaces is so weak that the attached organisms can be easily removed by the hydrodynamic shear force resulting from the ships' navigation or gentle mechanical cleaning. Among them, antifoulant-releasing coatings based on self-polishing copolymers (SPCs), which commonly are referred to as acrylate copolymers with hydrolyzable pendant groups, are the most commonly used ones.^{6,7} Tributyltin-based SPCs had been used for decades. However, they were prohibited in 2008 due to their serious ecological impact.⁸ Nowadays, tin-free self-polishing antifouling coatings containing silyl-, copper-, or zinc-based acrylate copolymers play the main roles in the market.^{9–12} In particular, silyl-based acrylate copolymers also exhibit self-smoothing properties and drag reduction, which can reduce energy consumption.¹³ SPCs were initially developed for ocean vessels that continue to travel, where the dissolution of the hydrolyzed polymer chain relies on the shear force of the strong water

flow.^{14,15} Hence, most SPCs exhibit a poor antifouling performance under static conditions when the fouling pressure is high. On the other hand, SPCs do not have fouling resistance ability themselves. Generally, antifoulants are added, which may cause ecological problems.

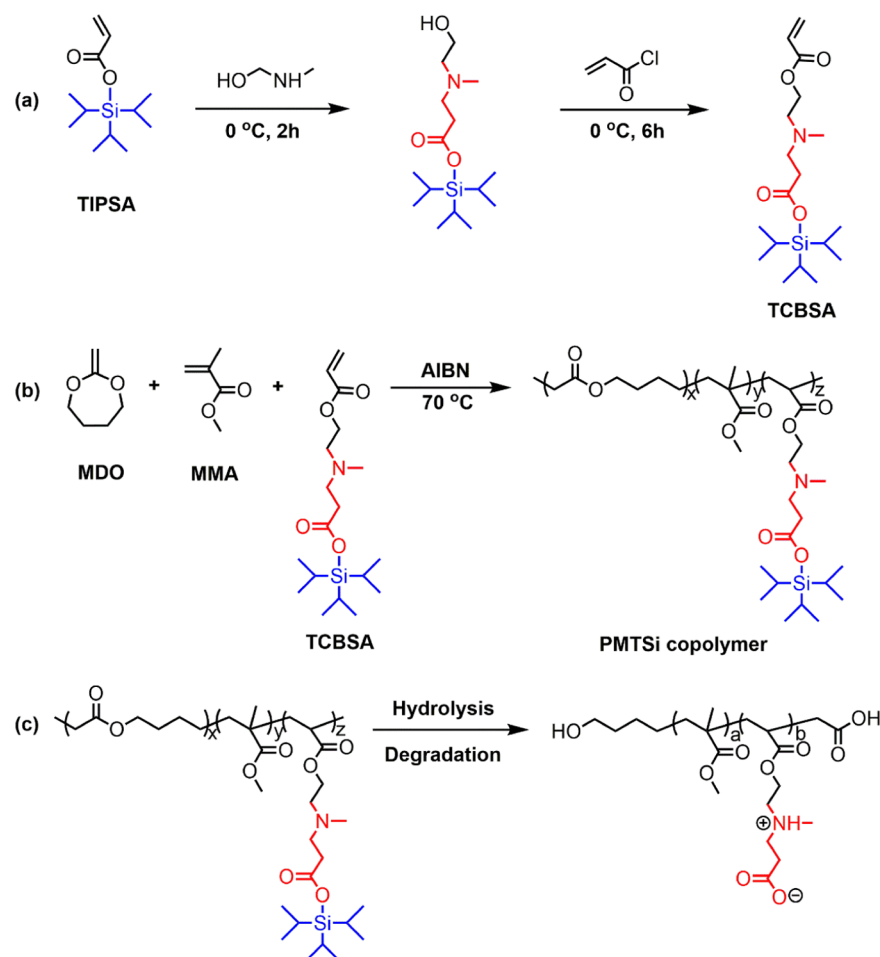
To improve the antifouling performance of SPC-based coatings, Bressy et al. developed diblock silyl-based copolymers by reversible addition–fragmentation chain transfer (RAFT) polymerization. Such diblock copolymers with well-defined microstructures have a well-controlled erosion rate and good antifouling performance compared with statistical copolymers.^{16,17} They also prepared block silyl-based copolymers with poly(dimethylsiloxane) (PDMS) by RAFT polymerization. The introduction of PDMS segments changes the surface wettability, and the coating can effectively resist the adhesion of hydrophilic marine bacteria.^{18,19} Recently, we developed a degradable SPC via the radical ring-opening copolymerization of 2-methylene-1,3-dioxepane (MDO), tributylsilyl methacrylate, and methyl methacrylate (MMA).²⁰ Such polymers can degrade via the cleavage of the main chain in marine environments, forming a self-renewing dynamic surface, and so the static antifouling performance is improved.^{20,21} However, the improvement is limited because

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Scheme 1. Synthesis of TCBSA and PMTSi15 and the Hydrolysis and Degradation of PMTSi15



a quick degradation also shortens the service life of the material. Generally, antifoulants have to be added to solve the contradiction.²²

On the other hand, it is well-known that poly(ethylene glycol) or zwitterions can resist the adsorption of proteins and microorganism. Some studies show that they can inhibit the adsorption of marine bacteria, algae, and diatoms.^{23–28} Therefore, the potential application of protein-resistant polymers for marine anti-biofouling has been examined.^{29–34} However, the mechanical strength of protein-resistant polymers is usually poor because the hydrophilic nature leads to the swelling of the matrix. Moreover, the complex marine environments prevent the protein-resistant surfaces from having a long-term antifouling performance since the surfaces cannot renew and are readily covered by some inorganics, thus losing their antifouling ability.^{29,34} Actually, the regeneration of protein-resistant polymer surfaces after contamination or damage is an important issue, especially in complex marine environments.

In the present study, based on the self-renewal of silyl acrylate polymers and the protein resistance of zwitterionic polymers, we synthesized a novel hydrophobic hydrolysis-induced zwitterionic monomer, tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSA), and copolymerized it with methyl methacrylate (MMA). Such self-generating zwitterionic coatings are expected to form a self-renewing surface with a persistent fouling resistance ability. Furthermore, TCBSA was copolymerized with MMA and 2-methylene-1,3-

dioxepane (MDO) by a facile radical ring-opening polymerization. The introduction of ester units into the main chain makes the polymers degradable, which further contributes to the self-renewal of the surface, especially under static conditions.²² We have investigated the surface wettability, hydrolytic and enzymatic degradation, protein resistance, and antifouling performance. We attempt to develop an ecofriendly marine anti-biofouling system.

RESULTS AND DISCUSSION

The synthesis routes of tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSA) and the typical copolymer with MDO, MMA, and TCBSA are shown in Scheme 1. To compare the effect of side groups on the rate of generation of zwitterions, we also prepared tertiary carboxybetaine ethyl ester ethyl acrylate (TCBEA). The ¹H NMR spectra of TCBSA and TCBEA are shown in Figures S1 and S2, respectively. For convenience, the copolymer containing 15 mol % MDO and 15 mol % TCBSA is named as PMTSi15, and its ¹H NMR spectrum is shown in Figure S3. Actually, the samples with different MDO or TCBSA contents have also been prepared to screen the optimal ratio. For a comparison, the copolymers of MMA (85 mol %) and tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSA, 15 mol %), tertiary carboxybetaine ethyl ester ethyl acrylate (TCBEA, 15 mol %), or triisopropylsilyl acrylate (TIPSA, 15 mol %) were synthesized and designated as PTSi15, PTE15,

Table 1. Characterization Data of Copolymers

entry	feed molar ratio MMA/MDO/hydrolyzable monomer	polymer composition ^a MMA/polyester/hydrolyzable units	$M_n \times 10^{-3}$ (g·mol ⁻¹) ^b	D_M ^b	T_g (°C) ^c
PMMA	100/0/0	100/0/0	22	1.9	113.0
PSi15	85/0/15(TIPSA)	85/0/15	27	2.3	66.0
PTE15	85/0/15(TCBEA)	85/0/15	27	2.1	57.1
PTSi15	85/0/15(TCBSA)	85/0/15	29	2.1	61.3
PMTSi15	56/32/12(TCBSA)	70/15/15	29	2.1	19.8

^aMolar percentage determined by ¹H NMR. ^bDetermined by GPC. ^cDetermined by DSC.

and PSi15, respectively. The characterization data of the copolymers are shown in Table 1.

Figure 1 shows the time dependence of the water contact angle (WCA) on the copolymer surfaces. All of the samples

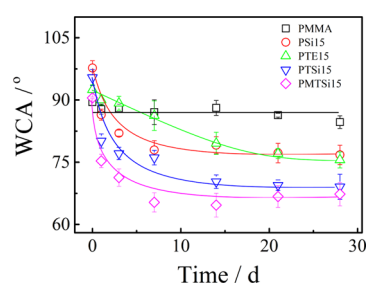


Figure 1. Time dependence of the water contact angle (WCA) of copolymer surfaces in ASW.

have an initial WCA > 90°, indicating that their surfaces are hydrophobic. After immersion in seawater for a certain time, the WCA of poly(methyl methacrylate) (PMMA) slightly changes since the linkage in PMMA is stable in ASW. For PSi15, the WCA sharply decreases from 98 to 78° in 7 days and levels off with the immersion time, indicating that the surface becomes more hydrophilic probably owing to the presence of carboxylate groups arising from the hydrolysis of triisopropylsilyl ester groups. For PTE15, the WCA gradually decreases from 93 to 76° in 28 days, indicating that the surface gradually becomes hydrophilic. The WCA of PTSi15 sharply decreases from about 95 to 70° in 14 days and reaches a constant WCA of ~67°. The WCA of PMTSi15 decreases more rapidly than that of PTE15, clearly indicating that the hydrolysis of tertiary carboxybetaine triisopropylsilyl ester is much faster than that of tertiary carboxybetaine ethyl ester, yielding much more zwitterions on the surface. For PMTSi15, the change of the WCA has a similar trend but has a much lower equilibrated WCA than that of PTSi15. This is probably because the introduction of polyester into the main chain makes the copolymer chain more flexible, and the tertiary carboxybetaine triisopropylsilyl ester can be readily attacked by seawater. Note that the roughness of these surfaces increases with the immersion time (Figure 4), and it could influence the WCA of the surfaces. However, all of the surfaces reach a constant WCA after 14 days even if the roughness continues to increase, indicating that the surface properties have much more effects on the WCA than the surface morphology in this system. Anyhow, the current data demonstrate that TCBSA-based copolymers can rapidly generate a hydrophilic zwitterionic surface by the induction of seawater.

The conversion process was further confirmed by attenuated total reflection infrared (ATR-IR) (Figure 2). For PMMA, the peaks in the spectrum have no changes since the chemical

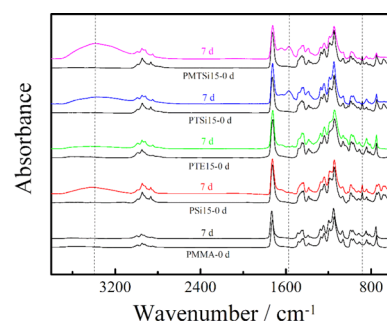


Figure 2. ATR-IR spectra of the copolymers with different immersion times in ASW.

structure of PMMA is stable in ASW. For PSi15 after immersion for 7 days, new peaks at 3387 and 1576 cm⁻¹ assigned to hydroxyl and carboxyl groups confirm the hydrolysis of triisopropylsilyl ester groups. In comparison, the peaks at 3387 and 1576 cm⁻¹ in PTE15 are not obvious, indicating the slow hydrolysis of tertiary carboxybetaine ethyl ester. However, significant peaks at 3387 and 1576 cm⁻¹ can be observed in PTSi15 or PMTSi15, indicating the rapid self-generated zwitterions. Meanwhile, the disappearance of the peak at 885 cm⁻¹ assigned to Si-C also confirms the hydrolysis of tertiary carboxybetaine triisopropylsilyl ester.

The mechanical strength is critical for antifouling coatings. Figure 3 and Table 2 show the adhesion strength and the

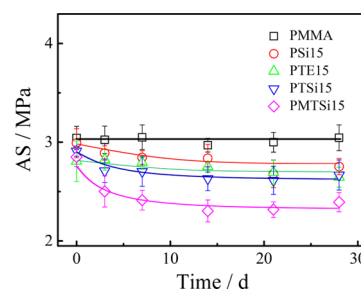


Figure 3. Adhesion strength (AS) of the copolymers to the substrate with different immersion times in ASW at 25 °C.

Table 2. Pencil Hardness of the Coatings with Different Immersion Times

time (days)	PMMA	PSi15	PTE15	PTSi15	PMTSi15
0	2H	2H	H	2H	H
3	2H	2H	H	H	HB
7	2H	2H	H	H	HB
14	2H	2H	H	H	HB
21	2H	2H	H	H	HB
28	2H	2H	H	H	HB

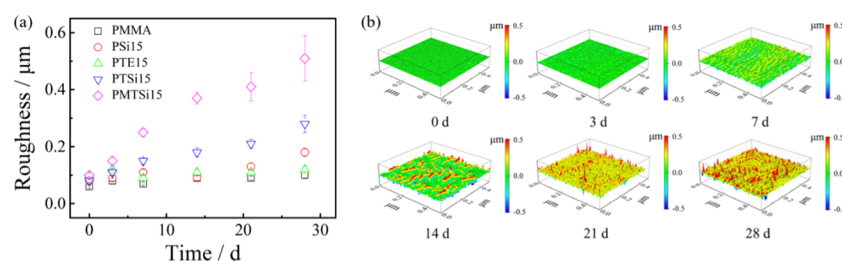


Figure 4. Surface roughness of the copolymers (a) and typical surface morphology of PMTSi15 (b) with different immersion times.

pencil hardness of the copolymers with different immersion times in ASW. Before immersion, the adhesion strength of each coating is above 2.7 MPa since the polymer mainly consists of acrylate segments, which has a high polar interaction and strong hydrophobic interaction with the substrate.^{35,36} Moreover, as shown in Table 1, the glass transition temperature of all polymers ranged from 20 to 113 °C, which is a suitable T_g for coatings.³⁷ Therefore, the acrylate segments and suitable T_g makes polymer coatings possible. After immersion, the adhesion strength of PMMA does not change, indicating that PMMA is stable in seawater. For PSi15 and PTE15, the adhesion strength slightly decreases. However, the decrease of adhesion strength for PTSi15 or PMTSi15 is somewhat obvious owing to the rapid hydrolysis of TCBSA. Anyhow, all of the samples have adhesion strengths above 2.0 MPa, and so the coatings are relatively stable even in harsh marine environments. A possible reason is that the hydrolysis of TCBSA occurs on the surface but the matrix is still hydrophobic. Thus, such polymeric coatings retain good mechanical strength during immersion. Similarly, the hardness grades of PMMA, PSi15, and PTE15 remain at 2H, 2H, and H, respectively. For PTSi15 and PMTSi15, the hardness grades decrease from 2H to H and from H to HB after immersion for 3 days but remain at H and HB, respectively, during the immersion. In other words, such polymeric coatings exhibit a stable mechanical strength in marine environments.

Figure 4 shows the surface roughness of the copolymers and the typical surface morphology of PMTSi15 with different immersion times. For PMMA, the surface roughness is kept at ~ 0.10 μm during the immersion since it is nonreactive in ASW. For PSi15, the surface roughness increases from ~ 0.10 to 0.15 μm after immersion for 28 days, indicating the occurrence of surface hydrolysis. For PTE15, the surface roughness slightly changes after immersion. As discussed above, the hydrolysis of the TCBEA group is so slow that the surface shows little changes in morphology. In contrast, the surface roughnesses of PTSi15 and PMTSi15 increase from ~ 0.10 to ~ 0.27 μm and ~ 0.10 to ~ 0.50 μm, respectively. This is because the faster hydrolysis of TCBSA leads to a much rougher surface.

Figure 5a shows the time dependence of mass loss of the copolymers in ASW at 25 °C. All of the samples except PMMA start to lose weight after immersion in ASW for 3 days and continue losing weight with the immersion time. For PMMA, the slight variation of mass loss is because the backbone is nonreactive in ASW. For PSi15, PTE15, and PTSi15, the mass losses after 28 days are ~ 6 , 3, and 10 wt %, respectively, which are attributed to the hydrolysis of side groups. The mass loss of PTSi15 is larger than that of PSi15, indicating that the hydrolysis of TCBSA is more rapid than that of TIPSA. PMTSi15 exhibits a much larger mass loss than PTSi15, indicating that the cleavage of the main chain ester can assist

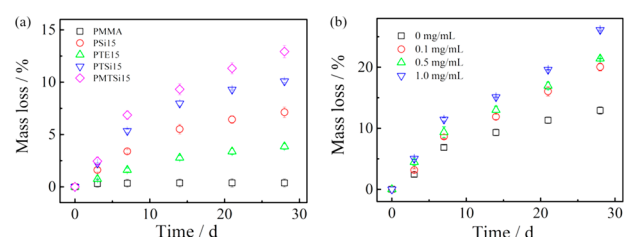


Figure 5. Time dependence of the mass loss of different copolymers (a) and the mass loss of PMTSi15 in different concentrations of lipase (b) in ASW at 25 °C.

the hydrolyzed zwitterionic segments to dissolve into ASW. In other words, the introduction of polyester segments can speed up the degradation. Note that the mass loss for a coating does not mean that it has undergone erosion since the mass loss can occur within the coating or in the bulk without any loss of thickness. Thus, we also measured the thickness loss of the copolymer (Figure S4). It has a trend similar to that of mass loss, clearly indicating that the copolymers erode after immersion in seawater. As reported before, adding MDO within the copolymer can easily adjust the hydrolysis of the polymer since the cleavage of the main chain ester can assist the hydrolyzed zwitterionic segments to dissolve into ASW. Moreover, it can decrease the swelling of coatings but maintain the self-polishing rate.^{18,20} We have also prepared copolymers with different MDO contents. As MDO contents increase, the mass loss increases (Figure S5). Therefore, the hydrolytic degradability can be tunable by the density of main chain esters besides the hydrolyzable side groups.

In the natural marine environment, various microorganisms can secrete enzymes and accelerate the degradation of polymers, which is conducive to the formation of dynamic surfaces to prevent the adhesion of marine organisms.³⁸ It is reported that degradable polymers such as aliphatic polyesters exhibit a faster mass loss in natural seawater than in artificial seawater owing to enzymatic degradation.^{39,40} However, the exact concentration of enzymes cannot be determined since the marine environment is complicated and changeable. Thus, we used a series of enzyme concentrations ranging from 0 to 1 mg mL⁻¹ to examine the enzymatic degradation of PMTSi15 in ASW at 25 °C (Figure 5b). Without lipase, the weight of PMTSi15 still decreases owing to the hydrolysis of side groups and degradation, and the mass loss of PMTSi15 is $\sim 12\%$ after immersion in ASW for 28 days. At the concentrations of 0.1, 0.5, and 1.0 mg mL⁻¹, the mass losses after 28 days are ~ 20 , 22, and 26%, respectively. Namely, as the lipase concentration increases, the mass loss of PMTSi15 increases. The result clearly indicates that the degradation of PMTSi15 is sensitive to the concentration of enzymes. As reported before, the process of biofouling involves the adsorption of micro-

organisms, and most of them can secrete enzymes.^{41,42} Therefore, the degradation of PMTSi15 has a fouling pressure response. Note that the current data only reflect that the copolymer can be enzymatically degraded and is sensitive to the concentration of enzymes. It cannot represent enzyme concentrations measured in the field.

Protein resistance is critical for marine antifouling since the initial process of biofouling involves the adsorption of protein and some fouling organics can secrete protein-based adhesives to colonize on submersed surfaces.⁴³ We evaluated the adsorption of proteins onto the copolymer surfaces with different immersion times using QCM-D. We used fibrinogen as the model protein since fibrinogen is a high-molecular-weight hydrophobic protein and it can strongly adsorb onto hydrophobic surfaces.⁴⁴ Figure 6 shows the time dependence

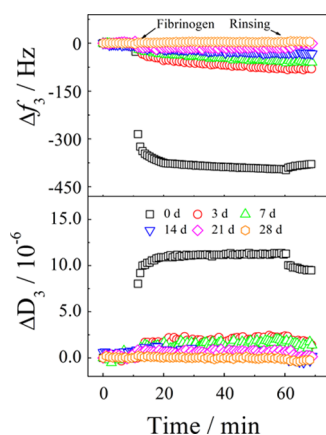


Figure 6. Time dependence of the frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of fibrinogen on PMTSi15 with different immersion times in ASW at 25 °C.

of the frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of fibrinogen onto PMTSi15 after different immersion times in ASW. Before immersion (0 day), as fibrinogen is introduced, the Δf decreases and ΔD increases sharply and subsequently levels off. An obvious decrease in Δf and an obvious increase in ΔD can still be observed after rinsing, indicating that the protein adsorbs onto the non-hydrolyzed surface forming a viscoelastic layer. As the immersion time increases, smaller changes in Δf and ΔD can be observed, indicating a decreasing protein adsorption. After immersion for 14 days, Δf and ΔD were close to the baseline, indicating that almost no protein is adsorbed. As discussed above, the side groups (TCBSA) in PMTSi15 can gradually convert to zwitterionic groups after hydrolysis, yielding a protein-resistant surface. Therefore, the adsorption of proteins decreases with the immersion time.

The final changes of frequency shifts (Δf) for different samples are summarized in Figure 7. For PMMA, the decrease of Δf is ~ 450 Hz and remains constant with different immersion times, indicating that the PMMA surface has no antifouling ability either before or after immersion in ASW. For PSi15, the decrease of Δf changes from ~ 410 to ~ 340 Hz after immersion for 3 days and levels off at ~ 300 Hz after 14 days, indicating that the protein resistance ability slightly increases. As reported before, triisopropylsilyl ester side groups can hydrolyze into anionic carboxylate groups, making the surface hydrophilic, which can decrease the protein adsorption to some extent.⁴⁵ For PTE15, the decrease of Δf changes from

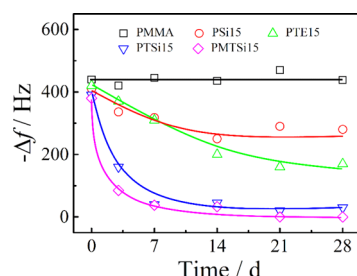


Figure 7. Frequency shift (Δf) regarding the adsorption of fibrinogen on copolymer surfaces with different immersion times at 25 °C.

~ 440 to 160 Hz with the immersion time, indicating that less proteins adsorb on the surface. Interestingly, the change of Δf for PTSi15 sharply decreases after 3 days of immersion and reaches ~ 30 Hz after 21 days, indicating the rapid generation of the protein-resistant surface. The difference between PTE15 and PTSi15 is the side groups. The sharp decrease of Δf in PTSi15 indicates that the side groups have great effects on the self-generating ability of zwitterions. In other words, the hydrolysis of TCBSA side groups is quicker than that of TCBEA and generates more zwitterions on the PTSi15 surface. For PMTSi15, the change of Δf is similar to that of PTSi15, but reaches ~ 0 Hz more rapidly. It is understandable that PMTSi15 has a flexible chain and the TCBSA side groups can be readily attacked by seawater. Namely, the introduction of polyester into the main chain accelerates the hydrolysis of TCBSA. Anyhow, the TCBSA-based coatings can rapidly self-generate zwitterionic surfaces with excellent protein resistance.

Figure 8 shows the fluorescence microscopy photographs of copolymer surfaces after immersion in a suspension of the marine bacteria *Pseudomonas* sp. and the relative bacterial adsorption (RBA) on the surfaces. For PMMA, a large quantity of bacteria can be observed on the surfaces with and without prehydrolysis, indicating that PMMA is susceptible to the adhesion of bacteria. For PSi15 without prehydrolysis, a large quantity of bacteria (RBA $\sim 87\%$) can be observed on the surface. After prehydrolysis in ASW for 3 days, there are still many bacteria (RBA $\sim 85\%$) adhered on the surface, indicating that the surface cannot effectively resist the adhesion of bacteria. PTE15 has a behavior similar to that of PSi15. A large number of bacteria (RBA $\sim 90\%$) are adhered on the surface without prehydrolysis. Even after prehydrolysis for 3 days, there is no obvious decrease (RBA $\sim 85\%$), indicating that PTE15 has a limited antibacterial performance. This is because there are not enough hydrolysis-induced zwitterions on the surface since the hydrolysis of TCBEA is slow. In contrast, almost no bacteria are adhered on either PTSi15 or PMTSi15 surfaces with prehydrolysis, indicating that they have generated enough zwitterions on the surface, which effectively resists the adhesion of bacteria. A small number of bacteria are adhered on PTSi15 (RBA $\sim 23\%$) or PMTSi15 (RBA $\sim 8\%$) surfaces even without prehydrolysis. This is probably because PTSi15 or PMTSi15 has generated some zwitterions during immersion in the bacterial suspension owing to the rapid hydrolysis of TCBSA.

Diatoms are a representative fouling organism that can easily colonize on submerged surfaces.⁴⁶ We have examined the anti-diatom performance of these copolymers (Figure 9). For PMMA, a great density of diatoms (~ 330 cell·mm⁻²) can be observed on the surface with and without prehydrolysis, indicating that PMMA is easy to be adhered by diatoms. For

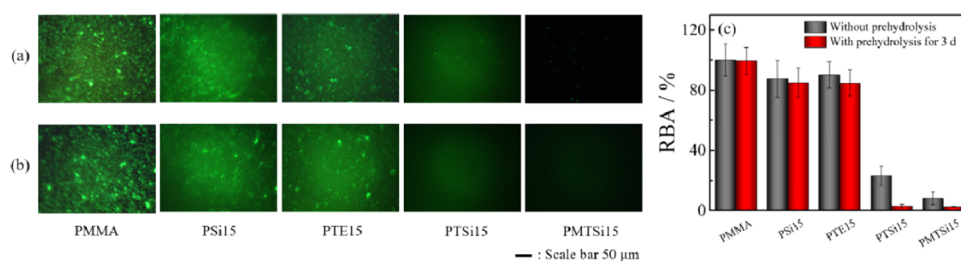


Figure 8. Fluorescence photographs of *Pseudomonas* sp. adhered on copolymer surfaces without (a) and with prehydrolysis for 3 days (b), and the relative bacterial adsorption (RBA) (c).

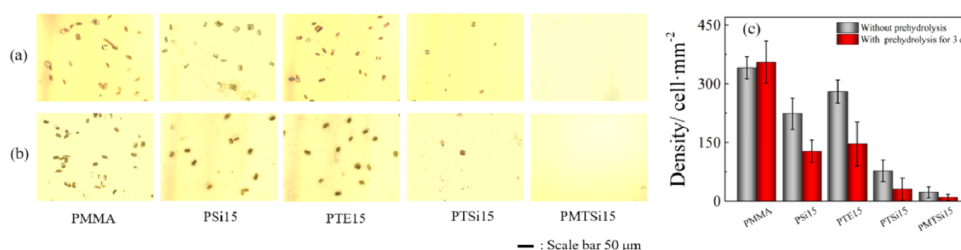


Figure 9. Photograph of the diatom *Navicula incerta* on copolymer surfaces without (a) and with prehydrolysis for 3 days (b) after 24 h of immersion in a diatom *N. incerta* suspension, and the density of diatoms on the surfaces (c).

PSi15 without prehydrolysis, a large number of diatoms (~ 230 cell·mm⁻²) settle on the surface. After prehydrolysis in ASW for 3 days, diatoms still settle on the surface but the quantity becomes smaller (~ 140 cell·mm⁻²). As reported before, diatoms tend to settle on hydrophobic surfaces.⁴⁷ Thus, the decreased density of diatoms on PSi15 with and without prehydrolysis can be attributed to the change of surface hydrophility. For PTE15, the densities of diatoms settled on the surface without and with prehydrolysis are ~ 290 cell·mm⁻² and ~ 150 cell·mm⁻², respectively, indicating that PTE15 also has a limited anti-diatom performance. In contrast, almost no diatoms settle on either PTSi15 or PMTSi15 surfaces with prehydrolysis, further indicating that the quick hydrolysis of TCBSA can induce enough zwitterions on the surface. Similar to the above antibacterial result, a low density of diatoms settles on PTSi15 or PMTSi15 surfaces without prehydrolysis because some zwitterions are generated during immersion in the diatom suspension. Anyhow, both antibacterial and anti-diatom results indicate that TCBSA-based coatings can rapidly generate hydrolysis-induced zwitterionic surfaces. Note that the rapid hydrolysis of TCBSA could shorten the shelf-life of coatings. Designing different substituent groups in tertiary carboxybetaine derivatives or optimizing the copolymer composition to adjust the hydrolysis rate is an important direction. Moreover, such coatings exhibit the fouling resistance property only after hydrolysis; the incorporation of functional substituent groups to make the coatings develop antifouling properties before hydrolysis is another future development of this technology.

CONCLUSIONS

Self-regenerating antifouling polymer surfaces have been developed by designing novel copolymers containing the hydrolysis-induced zwitterionic monomer tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSA). Such a copolymer rapidly self-generates a zwitterionic surface owing to its hydrolysis, and the hydrolyzed polymer chain can dissolve into seawater, leading to a self-renewing dynamic surface. Moreover, the introduction of ester units into the main

chain makes the polymers degradable, and the degradation rate is sensitive to the enzyme concentration, which further contributes to the self-renewing dynamic surfaces. They exhibit an excellent protein resistance and antifouling performance against marine bacteria *Pseudomonas* sp. and diatoms *N. incerta*. The coating with a self-generating and self-renewing zwitterionic surface is potential to find applications in marine anti-biofouling.

EXPERIMENTAL SECTION

Experimental details can be found in the [Supporting Information](#).

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the [ACS Publications website](#) at DOI: [10.1021/acsami.9b16775](https://doi.org/10.1021/acsami.9b16775).

Experimental section, ¹H NMR spectra of TCBSA, TCBEA, and PMTSi15, time dependence of the thickness loss of each polymer coating in ASW at 25 °C, and time dependence of the mass loss of PMxTSi15 in ASW at 25 °C ([PDF](#))

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Notes

The authors declare no competing financial interest.

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